

What Does a Yes/No LLM Recommender’s Confidence Know? Item Priors, Personal Evidence and Actionable Uncertainty

Anonymous Author(s)

Abstract

Pointwise LLM recommenders ask a yes/no question about a user-item pair and use the probability of *Yes* both to rank candidates and as a confidence for abstaining, exposure control and data pruning. What does that confidence know? We audit it for Qwen3-8B, zero-shot and LoRA-tuned (Llama-3.1-8B-Instruct as a replication backbone), on rated panels whose labels match the question and, zero-shot, on four same-candidate next-item panels (10k users, 101 candidates), in a pre-registered protocol with non-LLM references on the same pairs and confirmation on untouched users. We split the log-odds into a user offset, an item prior (the model’s mean answer for the item under other users’ histories) and a pair-specific residual, and measure how much the residual adds, as a predictor of the label, to an item mean and to matrix factorisation. Fine-tuned UAUC is below the item mean on all four rated panels (by 0.016 on ML-1M and, confirmed, by 0.08 to 0.13 on three Amazon domains), the item prior explains 66–90% of the within-user variance of the fine-tuned confidence, and the residual adds at most 0.010 UAUC to the prior and the item mean: positive only on ML-1M with Qwen3-8B, and not replicated with Llama. Fine-tuning mostly teaches the item: adapters trained on labels permuted within items or on item-mean pseudo-labels keep 85–92% of the gain (Qwen3-8B). **[DATA NEEDED: one sentence: which routes (cross-user selective serving against random and cheap signals, static exposure, item-dependent corrections, uncertainty-selected pruning, the method slot) beat their control, with effect size and CI, or that none did]** Because per-user recalibration cannot change within-user rankings, we test only the routes that can. All claims are scoped to same-candidate evaluation.

CCS Concepts

• Information systems → Recommender systems.

Keywords

LLM-based recommendation, confidence and calibration, item prior, popularity bias, pre-registration

1 Introduction

Large language models (LLMs) are increasingly used as *pointwise* recommenders: asked “Will this user like this item? Answer Yes or No,” the model’s probability of *Yes* is the ranking score [5, 69, 71]. The same number is read as a *confidence*: a system may abstain or truncate a list when it is low [12, 37], what is shown feeds back into what is learnt, a loop known to amplify popularity bias [9, 46], and fine-tuning data can be pruned by uncertainty [63] or coreset scores [43, 47]. Whether the confidence deserves this reading is

largely untested for the yes/no family, which is evaluated by discrimination (AUC, per-user AUC, UAUC) and log loss [42] rather than by reliability (calibration error, selective risk); uncertainty work on LLM recommenders addresses list-wise rankings [36, 66], generative recommenders [12] or verbalised confidence [54]. Output-level studies find that, under pointwise rating prompts, LLM recommenders follow item quality and user rating bias more than personal preference [57], that zero-shot LLMs trail collaborative filtering in rating prediction while fine-tuned ones approach it [32], and that LLM recommenders under-use the user’s history [33, 72]; none has measured what the token confidence itself encodes, how fine-tuning changes it, or which uses of it survive a matched control, on labels that match the question and against a leakage-free item mean and a collaborative-filtering reference on the same pairs [14].

Above all, it is open how much of $P(\text{Yes})$ is a prior about the item and how much is evidence about this user. Write the log-odds of *Yes* as $\ell(u, i) = b_u + \pi(i) + e(u, i)$: a user offset, an item prior that the model brings to the item whoever asks (possibly popularity-linked), and a pair-specific residual that carries whatever personal evidence there is. Only $\pi + e$ orders a user’s candidates, and the user offset is what per-user calibration cannot use: by an invariance lemma (Section 3) any strictly increasing per-user map, such as temperature or Platt scaling [21], leaves per-user AUC and within-user NDCG unchanged, so outcomes change only through item-dependent corrections, cross-user decisions and training. We measure personal information by prediction: the UAUC that the residual adds to the prior and an item mean, beside what matrix factorisation adds.

We audit the confidence of yes/no recommenders through six practitioner questions, S1–S6, posed as research questions with registered endpoints (Table 2): is the model sure when it is right (RQ1) and unsure when it is wrong (RQ2); does serving the most confident items over-expose head items (RQ3, static); how well does confidence track labels that match the question, and how much of it is item prior (RQ4); is it higher for popular items, and is that warranted once quality is controlled (RQ5); and does pruning the training examples that the zero-shot model is most uncertain about help more than matched random pruning (RQ6)?

A registered audit. Panels, endpoints, thresholds and a claim-admission rule were registered before any confirmatory result was analysed (Section 4): a statement beyond one panel needs confirmation on untouched users and the same sign, with its own interval excluding zero, on a fresh Amazon domain and on a second backbone. Not every gate was met: under the registered zero-shot prompt a pilot on 1,500 ML-1M users reached UAUC 0.587 against a bar of 0.60 (the next-item component was met: NDCG@10 0.2093 against 0.1863), and on the fully observed KuaiRec matrix [17] the zero-shot UAUC was 0.469 against 0.58, so exposure is analysed statically. One registered prompt-remedy round (Section 4) selected, on development users, the label-aligned prompt that asks whether

the user would rate the item 4 stars or higher; on 1,683 untouched ML-1M users it reached UAUC 0.603 (95% CI 0.594–0.612) against 0.576 for the registered prompt, so the bar (a point estimate) was met. As TALLRec-style recommenders are fine-tuned [5], the LoRA regime was to carry the main results if it met its own registered gate, a mean UAUC of at least 0.65 over three seeds. It did (mean 0.742; seeds 0.740, 0.745, 0.743), so the fine-tuned regime is the main object and the zero-shot regime its contrast.

Contributions.

- C1** *What the confidence encodes.* A decomposition of the yes/no log-odds into user offset, item prior and pair-specific residual, with the information gain over the prior-only item mean as the measure of personal information, benchmarked against an item mean and matrix factorisation on the same pairs; a within-user estimator and the complementarity to collaborative filtering; and three controls for what fine-tuning teaches: within-item label permutation, an item-only teacher and a store-name knockout with a real-brand placebo.
- C2** *Which uses of the confidence survive a control.* The invariance lemma and a controlled test of each route it leaves open: item-dependent corrections (including a two-view contrast [8] that we evaluate, not propose) against a paraphrase placebo and an ensemble null; selective serving [18] against random coverage; static exposure against the pool and eight published recommenders; uncertainty-selected pruning against class-matched random pruning over five seeds.
- C3** *A pre-registered, hash-chained protocol.* Qwen3-8B, zero-shot and with LoRA (three seeds), replicated with Llama-3.1-8B-Instruct, on rated panels with matched labels (ML-1M; Amazon Toys, Video Games, Sports) and, zero-shot, on same-candidate next-item panels of an earlier benchmark protocol [2] (four domains $\times$ 10k users $\times$ 101 candidates), with non-LLM references, user-clustered intervals, Holm correction within registered families, fresh-user confirmation and a table of deviations (Appendix A).

[DATA NEEDED: branch: only if the nested method slot (prior-offset LoRA) survives its kill rule – one sentence with its gain over post-hoc stacking, CI and datasets; else omit]

Findings in brief. (i) *What the confidence tracks* (Section 6.1). On the four rated panels (ML-1M, Toys, Video Games, Sports) the fine-tuned Qwen3-8B reaches UAUC 0.742, 0.558, 0.580 and 0.534, against 0.759, 0.684, 0.662 and 0.624 for the item mean on the same rows and 0.791, 0.539, 0.532 and 0.534 for matrix factorisation. It is below the item mean on all four panels, and Holm-confirmed below on the three Amazon panels (−0.126, −0.082, −0.090); on ML-1M the gap is −0.016 (exploratory), and Llama-3.1-8B matches the item mean there (0.760) and stays 0.123 below it on Toys. The item prior explains 0.90, 0.66, 0.72 and 0.74 of the within-user variance of the fine-tuned confidence (zero-shot: 0.39 on ML-1M and 0.33 on Sports; the other two panels fall below the reliability bar). The pair-specific residual adds at most +0.010 UAUC to the prior and the item mean (ML-1M, interval [0.003, 0.017]; at most +0.002 on the other panels, where the zero-shot Sports value, −0.009, is confirmed negative), against +0.036 for the MF residual on ML-1M; that supervision raises this gain (P1) holds on ML-1M (+0.011, $p = 0.004$, three

of three seeds) and finds no evidence on Llama-3.1-8B (+0.004, $p = 0.38$). (ii) *What fine-tuning teaches* (Section 6.2). An adapter trained on labels permuted within each item keeps $R = 0.92$ of the gain from real labels on ML-1M, and one trained on pseudo-labels from the item mean alone keeps 0.85 on ML-1M and, with a wide interval, 1.3 on Toys (both read ITEM_DRIVEN under the registered rule; the teacher adapters reach UAUC 0.720 and 0.564–0.571 against 0.742 and 0.558 for the real-label adapters). The fine-tuned LLM adds nothing detectable to matrix factorisation ($\mathcal{G}_{\text{LLM|CF}} \leq 0.005$, confirmed on none of the three Amazon panels), and the store-name knockout is INDETERMINATE on every panel and model, so the confidence is called popularity-linked, not familiarity-linked. **[DATA NEEDED: (iii) Which uses survive a matched control (Section 6.3): selective serving against random and against cheap signals, static exposure, item-dependent corrections, uncertainty-selected pruning (S6) and the method slot, to be written from the finished runs]** Table 6 condenses the answers to S1–S6.

2 Related Work

Pointwise yes/no scoring. TALLRec [5] tunes an LLM with LoRA to answer whether a user will like an item, CoLLM [71] and BinLLM [69] add collaborative information, and ReLLa [42] retrieves behaviours and scores a pair by the two-way Yes/No softmax, the sigmoid of the logit audited here, reporting AUC, log loss and accuracy. IR rankers score relevance by the probability of a “true” token [49]; finer relevance grades help zero-shot pointwise rankers [78], and pairwise prompting avoids the calibration across prompts that pointwise scores need [53]. Generative recommenders [3, 19, 40] face distorted sequence-level confidence [4], which one Yes/No position avoids; zero-shot LLM rankers for recommendation are swayed by popularity and candidate position [27], and LLMs partly memorise MovieLens-1M [11]. Unlike CoLLM and BinLLM, the optional prior-offset LoRA adds a stop-gradient item-prior offset to the logit. We ask what the yes/no confidence encodes, beside non-LLM references on the same pairs.

Confidence, calibration and answer priors. LLMs are overconfident when they verbalise confidence [65] but fairly calibrated in token probabilities on multiple-choice questions [31]. Binary answers carry input-independent priors, removed by contextual [75] or batch calibration [76] or PMI scoring [25]; yes/no answers show wording-driven bias with idiosyncratic per-item residuals [29], LLMs prefer recognised brands [10], and the two-view contrast MIRROR belongs to contrast-consistent search [8]. For LLM recommenders, UQRec [36] decomposes list-wise uncertainty, UGR [12] uses confidence tokens for rejection and truncation, EviRank [66] estimates position-level confidence, an audit scores verbalised confidence by popularity stratum [54], KnowSA [39] finds popularity and uncertainty poor proxies of item knowledge, FACTER thresholds a fairness score conformally [13], and selective escalation to an LLM did not help in a sparse study [77]. None asks how much of the token confidence is a user-independent item prior.

Popularity, exposure, pruning and serving. Popularity bias [1] has causal treatments [62, 70] and is amplified by SFT and DPO

in LLM recommenders [15, 16], whereas a prompted LLM recommender showed less of it than conventional ones [41]; in entity question answering, LLM confidence rises with answer popularity [48]. Feedback loops raise homogeneity and popularity bias [9, 46], also with LLM components [51]; we measure exposure statically. LLM-recommender training data are selected by influence and effort [43] or coresets [47], instruction tuning prunes at an uncertainty quantile [63], and recommenders are denoised by dropping large-loss interactions [59] or by thresholding an LLM’s generation probability [58]. Serving by a confidence score is a performance-prediction decision [6] and selective classification [18], which can widen group gaps [30]. We test each use against a matched control.

3 Setting and Notation

Yes/no recommender and token confidence. A pointwise yes/no recommender [5, 71] scores a user u and a candidate item i by asking a binary question q about the pair (default q = like, “Would this user like the candidate item?”) and reading the next-token distribution of a language model f , zero-shot or with a LoRA adapter fitted to yes/no labels. Its *confidence* is the log-odds

$$\ell(u, i) = \log P_f(\text{Yes} \mid x_q(u, i)) - \log P_f(\text{No} \mid x_q(u, i)), \quad (1)$$

where $P_f(\text{Yes} \mid x)$ sums the probabilities of the vocabulary ids that decode, stripped and lower-cased, to “yes” (likewise No); candidates are ranked by ℓ .

Panels and metrics. A *unit* is a user (rated panels) or a prediction event (next-item panels) with candidate set C_u and labels $y(u, i) \in \{0, 1\}$. *Rated panels* show the user’s earlier ratings as history and use the user’s own later rated items as candidates, labelled by whether the user liked them (Section 4). With $c(x) = 1, \frac{1}{2}, 0$ for $x > 0, x = 0, x < 0$, the per-user AUC of a score s and its mean over users with both classes (UAUC [69, 71]) are

$$\text{AUC}_u(s) = \frac{1}{|C_u^+||C_u^-|} \sum_{i \in C_u^+} \sum_{j \in C_u^-} c(s_i - s_j), \quad \text{UAUC} = \text{mean}_u \text{AUC}_u.$$

Same-candidate next-item panels hold, per event, one positive and 100 popularity-sampled negatives, ranked identically by every system (not full-catalogue ranking [35]). With a candidates above the positive and t tied with it, its rank R is uniform on $1+a, \dots, 1+a+t$, and $\text{HR}@k$, $\text{NDCG}@k$ and MRR are the expectations of $1[R \leq k]$, $1[R \leq k]/\log_2(1+R)$ and $1/R$ ($k=10$).

Decomposition and audit quantities. Let $\pi(i) = \mathbb{E}_D[\ell(D, i)]$ be the *item prior*, the expected confidence for i under the history of a donor D who has not seen i (estimated by the mean $\hat{\pi}(i)$ over $K = 8$ donors). With the user offset $b_u = |C_u|^{-1} \sum_{i \in C_u} [\ell(u, i) - \pi(i)]$, the remainder is the *pair-specific residual* e :

$$\ell(u, i) = b_u + \pi(i) + e(u, i), \quad \sum_{i \in C_u} e(u, i) = 0, \quad (2)$$

so within a unit only $\pi + e$ orders candidates. The residual e is not personal evidence by construction: it also holds donor-population mismatch (donors have not consumed i), user-by-prior interaction and history noise unrelated to the label. The audit therefore measures personal information by prediction. On rated panels it reports (estimators: Section 5) the *information gain* $\mathcal{G}$, the UAUC gain of cross-fitted stackers when $\hat{e} = \ell - \hat{\pi}$ is added to $\hat{\pi}$ and the prior-only item mean $m(i)$ (other users’ mean rating of i before

the candidate’s timestamp), beside the analogous gain $\mathcal{G}_{\text{CF}}$ of a matrix-factorisation residual over m ; the *non-prior share* $1 - \rho^2/r_8$, the share of the within-unit variance of ℓ that $\hat{\pi}$ does not explain linearly (ρ the pooled within-unit correlation of ℓ and $\hat{\pi}$, r_8 the Spearman–Brown reliability [7, 56] of the 8-donor mean), with the variance share of e itself beside it; and the *popularity link*, the partial Spearman correlation of ℓ and of $\hat{\pi}$ with log popularity given $m(i)$.

Where can confidence change a ranking?

LEMMA 3.1 (INVARIANCE). *If every $g_u : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing and $s'_i = g_u(s_i)$ for $i \in C_u$, then*

$$\text{sgn}(s'_i - s'_j) = \text{sgn}(s_i - s_j) \quad \text{for all } i, j \in C_u,$$

so AUC_u , $\text{HR}@k$, $\text{NDCG}@k$ and MRR , which depend on s only through these signs (via a , t and c), are unchanged under either tie convention.

Strict monotonicity preserves $>$, $=$ and $<$. Temperature [21] and Platt scaling [52] with positive slope, removal of b_u and the list softmax $w_i \propto \exp(\ell_i/T)$ are such maps: per-user calibration cannot change UAUC or within-unit NDCG. Merely non-decreasing maps (isotonic regression [68]) can, through ties.

PROPOSITION 3.2 (ITEM-DEPENDENT CORRECTIONS). *For continuous, strictly increasing $g_{u,i}$ and $s'_i = g_{u,i}(s_i)$, the weak order of C_u is unchanged for every score vector if and only if all $g_{u,i}$ coincide; for $s' = s - \lambda b(i)$, a pair with $s_i > s_j$ is exchanged or tied exactly when*

$$s_i - s_j \leq \lambda (b(i) - b(j)).$$

“If” is Lemma 3.1; if $g_{u,i}(x) < g_{u,j}(x)$, continuity gives $\varepsilon > 0$ with $g_{u,i}(x+\varepsilon) < g_{u,j}(x)$, so $s_i = x+\varepsilon > s_j = x$ is reversed. Contextual [75] and batch calibration [76] subtract a prior that, for a binary question, is a constant of the unit; candidate-conditioned priors (the no-history prompt, in the spirit of PMI scoring [25]; the swap prior $\hat{\pi}$) and second readouts of the pair (a mirrored or paraphrased question) are item-dependent, and as reordering is not improving each is compared with an equal-compute control. Across units the lemma is silent: offsets change pooled AUC, thresholds and selective serving, and training changes f .

4 Audit Protocol

Registration. Three dated, binding amendments with addenda register the audit (Appendix A), each hashed before the runs it governs (Amendment 1 by its commit and pull times): operationalisations before any pilot data, the prompt remedy after the pilot gates, and the fine-tuned programme before any adapter existed. The next-item panels were scored under a registered carve-out and analysed by a specification frozen before any audit score was analysed. Later addenda add the endpoints marked † , the item-only teacher and the wording rules below; Appendix A lists every deviation and post-hoc choice. Quantities computed on design users are *exploratory*.

Panels, labels and user roles. *Rated panels* (MovieLens-1M [22]; Amazon Reviews 2023 [26] Toys, Video Games, Sports) set $y = 1$ for rating ≥ 4 and $y = 0$ for ≤ 2 ; a user needs three history events and, among at most 20 candidates (the latest rated items), three likes and three dislikes. Eligible users are shuffled with seed 0: the first 1,500 are TRAIN users (on ML-1M and Toys the pilot users, i.e. the DEV users of the remedy), the next $\min(3,000, \text{eligible} - 1,500)$

EVAL users (on ML-1M the 1,683 CONFIRM users of the gates), disjoint and never trained or selected on. With T_d the 0.8-quantile of a panel’s candidate timestamps, training uses TRAIN candidates before T_d , calibrators EVAL candidates before T_d (CAL), and every rated endpoint but the G6 gate EVAL candidates at or after T_d (TEST). *Next-item panels* are the frozen same-candidate panels of an earlier benchmark protocol [2] (Amazon Sports, Toys, Home, Tools; 10,000 test events, each with one positive and 100 popularity-sampled negatives), with the top-10 lists of a verbalised reranker and eight published recommenders; 2,000 VALID events fit temperatures only, and Sports TEST events 1–1,000 are quarantined and reported separately.

Backbones, regimes and readout. The primary backbone is Qwen3-8B [67]; Llama-3.1-8B-Instruct [20] replicates on ML-1M and Toys (both regimes) and on next-item TEST events 1,001–3,000 (zero-shot), subject to registered cut rules, and never rescues a Qwen result. Rated panels share the prompt chosen by the remedy round, next-item panels keep the registered prompt V0, and no adapter is scored on them. Scoring is one fp16 prefill pass in vLLM [38] reading Yes and No by token id; a run that fails the integrity check E1 (at most 0.5% of rows without an answer token, no over-length prompt, mean Yes+No mass ≥ 0.95) is rerun once and otherwise reported missing, never replaced. The fine-tuned regime is TALLRec-style [5] LoRA [28] yes/no SFT with one fixed recipe (seeds 0–2, five for pruning; Appendix A).

Registered gates and the remedy round. The bars are an ML-1M zero-shot rated UAUC of 0.60 (point estimate), a Sports next-item NDCG@10 of 0.1863 (0.8 times the verbalised reranker) and a Gate-FT mean UAUC of 0.65. Under V0 the zero-shot pilot missed the rated bar and met the next-item bar, and the KuaiRec pilot [17] was NULL, so no fully-observed-label or dynamic claim is made. One prompt-fix round, blind to MIRROR and with the bars unchanged, scored V0 and six variants on the DEV users (Appendix A), found a fix (FIX_FOUND) and confirmed it once on fresh ML-1M users (GATE_PASS on the point estimate; interval in Table 1). Gate-FT (LoRA on the DEV users’ ML-1M candidates before T_d , scored on the CONFIRM users’ candidates after it) passed (GATE_FT_PASS); its bar lies between published ML-1M UAUCs of MF (0.636) and fine-tuned TALLRec (0.682) with 3-star negatives [71], so it licenses the fine-tuned programme, not competence, which Table 3A judges against the item mean and MF on the same rows.

Table 1: Registered gates of the remedy round and Gate-FT (DEV: 1,500 users per panel; CONFIRM: 1,683 fresh ML-1M users, Gate-FT on their candidates after T_d ; the pilot outcomes are in the text).

Gate (bar)	V0	V* / LoRA	Outcome
G5 DEV: ML-1M UAUC (bound > 0)	0.588	0.612	FIX_FOUND
G5 DEV: Toys UAUC (E2)	0.540	0.544	yes
G6 CONFIRM: ML-1M UAUC (≥ 0.60)	0.576 $\pm$ 0.009	0.603 $\pm$ 0.009	GATE_PASS
G9 Gate-FT: seeds 0 / 1 / 2	–	0.740 / 0.745 / 0.743	
G9 Gate-FT: mean (≥ 0.65)	–	0.742 $\pm$ 0.019 (.002)	GATE_FT_PASS

Non-LLM references. Rated tables carry on identical rows, as context and never as gates, the prior-only item mean $m(i)$ (Section 3; shrunk with $k = 5$), which may use other users’ ratings in $[T_d, t)$

that the LLMs never see, so the information-matched mean $m_T(i)$ (ratings before T_d only) stands beside it; and a temporal biased matrix factorisation (MF) [34] fitted before T_d , whose personal residual (MF – $b_u - b_i$) is the CF personal-signal reference. Next-item tables carry the verbalised reranker and the eight published recommenders [23, 44, 45, 55, 60, 61, 73, 74] as context, never as a state-of-the-art claim; every claim is a same-candidate claim.

Statistics, families and claim admission. Intervals are 95% percentile bootstraps (2,000 resamples, seed 0) over users, events or, for item-level correlations, items; contrasts are paired and p -values two-sided. Holm’s correction [24] applies within each registered family: E-B, the regime contrast over the four Qwen domains; E-D, $\mathcal{G}$ and the star-permutation contrast per domain and regime; the method slot over its datasets; each next-item endpoint over the four domains; and, on the three Qwen Amazon panels, E-F ($\mathcal{G}_{\text{LLM}|CF} > 0$, fine-tuned), E-H ($\mathcal{G}_{\text{prior}} > 0$ on sparse rows, per regime) and E-J (two-sided: UAUC of ℓ minus that of m , fine-tuned). P1 and the pruning contrast P2–P1 are single tests at $\alpha = 0.05$. A fine-tuned result is confirmed only if, beyond Holm $p < 0.05$, its seed mean exceeds $2\sigma_{\text{seed}}$ with all seeds of one sign; an endpoint on fewer than 150 users is descriptive. A direction word is written only where such a test decides it, and a family with mixed outcomes is reported as counts per regime (“confirmed in k of 4 zero-shot panels”). A statement on personal information gain is *robust* only if E-D and the within-user estimator E-W agree in sign with intervals excluding 0 (fine-tuned: all seeds), else estimator-dependent. A finding enters the abstract only if it is confirmed on fresh users and *replicates*: the same sign with its own unadjusted interval excluding 0 on a fresh Amazon domain and on the second backbone (Llama’s rated arms or next-item sample; pruning and the method slot qualify only with a Llama replication). No re-anchoring, near-miss rule or arm promotion is allowed, and a configuration matching no registered branch is INDETERMINATE. Table 2 maps S1–S6 to RQ1–RQ6.

5 Measurement and Actionability Tools

Calibration and error anatomy. On rated panels a global Platt map [52] of the logit, fitted on all CAL rows of a model’s EVAL users, is applied to TEST (ECE with 10 equal-mass bins [50]). For a user with both classes among the TEST rows the *oracle* top- k_u decision sets k_u to the user’s TEST likes and c_u to the mid-point of the k_u -th and (k_u+1) -th largest TEST logits and predicts $1[\ell \geq c_u]$ with margin $|\ell - c_u|$; the *deployable* decision[†] uses $k_u = \text{clip}(\text{round}(r_u n_u), 1, n_u - 1)$, with n_u TEST rows and r_u the user’s like rate among CAL rows (pooled below 3 CAL rows). Over margin tertiles of all TEST pairs we report the shares of errors in the top and of correct decisions in the bottom tertile, and the margin’s AUROC for correctness.

Prior, shares and information gain. The swap prior $\hat{\pi}(i)$ averages $\ell_{\text{like}}(d, i)$ over $K = 8$ donors d who hold i neither in their history nor as a candidate, drawn from S_d , the EVAL users with both classes among their TEST candidates (the first 1,000; here all); an empty-history prompt gives the no-history logit $\ell_0(i)$ [25]. With r_c the pooled within-user correlation of the centred donor-half means, $r_8 = 2r_c/(1+r_c)$ and ρ the pooled within-user correlation of centred ℓ and $\hat{\pi}$, the *item-prior share* is ρ^2/r_8 and the non-prior share its

Table 2: Research questions (RQ k answers seed question S k) and registered endpoints (Qwen3-8B, both regimes; Llama-3.1-8B replication): rated on TEST candidates of EVAL users, next-item on TEST events; p : list-normalised probability; † : addenda 6–8.

RQ	Rated panels	Next-item panels	Rule
RQ1: sure when right?	margin AUROC and AURC for the correctness of the per-user top- k_u decision (oracle; deployable †); correct decisions in the bottom margin tertile	AUROC of $p_{\max}$ for top-1 correctness; accuracy in the bottom and top $p_{\max}$ tertile	next-item: Holm over the four domains
RQ2: unsure when wrong?	errors in the top margin tertile	top-1 errors in the top tertile	as RQ1
RQ3: over-exposure?	—	Δ_{head} : Gini, APLT; NDCG@10 gain and niche served share at 50% coverage vs. random	Δ_{head} : one sign, interval excluding 0, in ≥ 3 of 4 domains; static only
RQ4: what is tracked?	UAUC beside m , $m_T^\dagger$ and MF (E-A, E-J †); regime contrast (E-B); ECE, Platt slope; item-prior share, $e\text{-share}^\dagger$; $\mathcal{G}$ (E-D) and $\mathcal{G}_{\text{wu}}$ (E-W †) beside $\mathcal{G}_{\text{CF}}$; E-F † ; strata (E-H †); star permutation; FT-C, FT-Q retention †	per-event AUC; top-label ECE, Brier; NDCG@10	$r_8 < 0.7$: shares uninterpretable; P1 (ML-1M): mean $\mathcal{G}_{\text{FT}} - \mathcal{G}_{\text{ZS}} > 0$, $p < 0.05$, 3/3 seeds, $> 2\sigma_{\text{seed}}$; H-F, H-S, H-J † ; “teaches the item” needs FT-C (ML-1M) and FT-Q (each dataset run) ITEM_DRIVEN †
RQ5: popular, so confident?	popularity link of ℓ and $\hat{\pi}$ given m ; head–tail Bias Index; store-name knockout	mean p , residual $y - p$ and Bias Index by popularity group	knockout POSITIVE: head–tail $\Delta_{\text{pseudo}} \geq 0.20$ (interval above 0), $ \Delta_{\text{placebo}} \leq 0.05$, $ \Delta\text{UAUC}_{\text{tail}} \leq 0.01$; “familiarity” only if every Qwen model on Toys and Video Games is POSITIVE and Llama on Toys agrees
RQ6: does pruning help?	ML-1M TEST UAUC after removing 25% of each label class at random (P1), uncertainty-selected (P2) or prior-congruent (P3); full data (P0); seeds 0–4	—	P2–P1 beats iff interval above 0, mean > 2 SD of paired differences, 5/5 positive; equal iff interval within ± 0.01 ; arms match classes, not items (diagnostics; after a win, an item-stratified arm †)

complement (uninterpretable if $r_8 < 0.7$). The $e\text{-share}^\dagger$, $[\text{Var}(\ell_c - \hat{\pi}_c) - (1 - r_8)\text{Var}(\hat{\pi}_c)]/\text{Var}(\ell_c)$ over user-centred values, removes the donor-mean error from the variance share of $\hat{e}$; the item shares of the MF score (its item bias b_i) and of the label (m , a lower bound) are computed alike † . Logistic stackers are cross-fitted over two user halves of S_d : M0 uses m , M1 adds $\hat{\pi}$, M2 adds $\hat{e} = \ell - \hat{\pi}$, M3 pairs m with the MF personal residual, and M4 † joins M2 and M3; $\mathcal{G} = \Delta\text{UAUC}(\text{M2} - \text{M1})$, $\mathcal{G}_{\text{CF}} = \Delta\text{UAUC}(\text{M3} - \text{M0})$, $\mathcal{G}_{\text{LLM}|\text{CF}} = \Delta\text{UAUC}(\text{M4} - \text{M3})$ and $\mathcal{G}_{\text{CF}|\text{LLM}} = \Delta\text{UAUC}(\text{M4} - \text{M2})$. E-D fits the stackers on raw features, as registered; as user offsets that track the labels can tilt such weights, E-W † refits them on user-centred features over 20 user-half splits (one is E-D’s), averages each user’s ΔUAUC over splits and resamples users with the coefficients fixed ($\mathcal{G}_{\text{wu}}$, $\mathcal{G}_{\text{CF},\text{wu}}$, P1 $_{\text{wu}}$; E-F and E-H use E-W). The star permutation shuffles the “(rated $r/5$)” suffixes among the history events (two permutations); the model is said to ignore the ratings if $|\Delta\text{UAUC}| < 0.005$ and the within-user s.d. of $\ell - \hat{\ell}_{\text{perm}}$ is below 0.10.

Strata, matched mean and item-only controls. E-H † splits TEST rows into *sparse* (m from fewer than 5 earlier ratings) and dense rows and into *unseen* (no TRAIN example) and seen items, and evaluates $\mathcal{G}_{\text{prior}} = \Delta\text{UAUC}(\text{M1} - \text{M0})$ per stratum with predictors fitted on all rows. E-J † pairs ℓ with m , with m_T and, on warm pairs, with MF. Two item-only controls train two adapters each (seeds 0, 1, same recipe) without user–item evidence: FT-C on labels permuted within each item (ML-1M only: a permutation cannot change an item with one TRAIN example) and FT-Q on the labels of an item-only teacher, positive for the round(βn) TRAIN examples with the largest m (β : TRAIN like rate), so it uses no label of the example itself, no user information and nothing after T_d . The retention R (FT-Q: R_Q) averages $[\text{UAUC}(p_s) - \text{UAUC}_{\text{ZS}}]/[\text{UAUC}(s) - \text{UAUC}_{\text{ZS}}]$ over the adapters s and their control twins p_s , is defined when the regime contrast is positive with an interval excluding 0, and reads † ITEM_DRIVEN if its interval lies above 0.5, USER_DRIVEN if below, MIXED otherwise.

Popularity and the store-name knockout. Popularity n_i is the all-time category event count, used through ranks. *Level*: the partial Spearman correlation of ℓ and of $\hat{\pi}$ with $\log(1 + n_i)$ given $m(i)$

(and release year on ML-1M, description length and a has-store flag on Toys and Sports). *Gap*, following ProCal [64]: with confidence c , residual $r = y - c$ and ten equal-mass confidence bins b , $\text{BI}_g = \sum_b n_{gb}(\bar{r}_{gb} - \bar{r}_b)/\sum_b n_{gb}$, reported as $\text{BI}_{\text{head}} - \text{BI}_{\text{tail}}$. In the knockout, fixed models (zero-shot and every adapter; Qwen on Toys and Video Games, Sports unless cut; Llama on Toys) score the EVAL panel with every store-name string (the Amazon store field, wherever it occurs in titles and texts) replaced by an invented pseudonym, and in a placebo arm by another real store of the same review-count tertile; sub-brands, characters and product lines remain. On candidates whose store string occurs in their title, $\Delta_{\text{arm}} = \ell_{\text{real}} - \ell_{\text{arm}}$ is summarised by its head-minus-tail contrast and the tail ΔUAUC , and a registered decision function labels each model (Table 2; Appendix A).

List-normalised confidence, serving and exposure. On next-item panels the 101 logits of an event are normalised to $p_c \propto \exp(\ell_c/T)$, with T fitted per domain, question and backbone on VALID events (rank-inert within an event, Lemma 3.1); top-label ECE (15 equal-width bins) and the Brier score use $p_{\max} = \max_c p_c$ against top-1 correctness. Selective serving [18] serves events by decreasing $p_{\max}$ (random control) and at 50% coverage reports the NDCG@10 gain and the served shares of niche and mainstream users (lowest and highest non-empty quintile of a profile averaging head 2, mid 1, tail 0 over history items). Exposure is read from the top-10 lists: tail share (APLT), the Gini coefficient of exposure counts, and Δ_{head} , the head share minus its expectation under random ranking of the same candidates.

Item-dependent corrections and MIRROR. Per-user Platt scaling and offset removal check Lemma 3.1; the routes of Proposition 3.2, the no-history correction $\ell - \ell_0(i)$ and the residual $\hat{e}$, are compared with the raw score. MIRROR, $\ell_{\text{like}} - \ell_{\text{dislike}}$, is scored only after GATE_PASS and must beat the equal-compute paraphrase placebo $\frac{1}{2}(\ell_{\text{like}} + \ell_{\text{like_para}})$ and a correlation-matched ensemble null, losing at most 0.005 NDCG@10 on 1,000 Sports VALID events.

Pruning arms (RQ6). On ML-1M, P0 trains on all TRAIN examples and each pruned arm removes 25% of the positives and 25% of the negatives: P1 at random, P2 the zero-shot model’s most uncertain

examples (largest $-|\ell_{ZS} - \tau|$, τ the $(1 - \beta)$ -quantile of its logits and β the TRAIN label mean; in the spirit of UNIT [63]), P3 the most prior-congruent, by $(2y - 1)(m - \bar{m})$; seeds 0–4. P1 matches label classes, not items. A win reads “pruning the zero-shot model’s uncertain examples helped”, never “uncertainty improves ranking”.

Optional method: prior-offset LoRA. The adapter can be made to model only what the prior does not: in training the Yes logit is shifted by $b z(m)$ before the softmax ($z(m)$ the TRAIN-standardised item mean, b a learned scalar initialised at 0, no gradient into m), and the score is $\ell + b z(m)$. It is compared, on the same seeds and TRAIN set, with post-hoc stacking of the SFT logit and $z(m)$ fitted on CAL; a dataset passes if the seed-averaged ΔUAUC is at least $+0.01$, its interval excludes 0 and the $2\sigma_{\text{seed}}$ rule holds, and the slot is killed after two failures (order ML-1M, Toys, Video Games, Sports; at the latest on 2026-11-30).

6 Findings

Every number is read from a committed result file: cells give the estimate $\pm$ the half-width of its 95% bootstrap interval over users or events (LoRA: seed mean, interval and seed s.d.; “(descriptive)”: fewer than 150 users), a direction is written only where a registered test decides it, and registered outputs that no table shows are released with the artefact.

6.1 What the confidence tracks

Table 3A sets every LLM UAUC beside the item mean, the information-matched item mean and MF on the same rows. Fine-tuning raises UAUC (E-B, confirmed on 4 of 4 Qwen panels), and in the fine-tuned regime the UAUC of ℓ differs from that of the item mean (H-J) on 3 of 3 Qwen Amazon panels (below it on Toys, Video Games and Sports). Table 3B gives the share of the within-user variance of the confidence that the item prior explains (the non-prior share is its complement), beside the e-share and the item shares of MF and of the labels. The personal information gain $\mathcal{G}$ is confirmed (E-D) on 1 of 4 (negative: Sports) zero-shot and 1 of 4 (positive: ML-1M) fine-tuned panels, beside the CF reference gain $\mathcal{G}_{\text{CF}}$, and the within-user estimator finds it robust on 0 of 3 zero-shot and 0 of 3 fine-tuned Qwen Amazon panels; estimator-dependent on Video Games (zero-shot), Sports (zero-shot), Video Games (fine-tuned). P1 holds (mean $\mathcal{G}_{\text{FT}} - \mathcal{G}_{\text{ZS}}$ 0.011 $\pm$ 0.008, p 0.004; 3 of 3 seeds positive; within-user reading robust (exploratory)), so **[DATA NEEDED: branch: supervision raises the personal information gain / there is no evidence that supervision adds personal evidence]**. Wording flags: shares need $r_8 \geq 0.7$, $\mathcal{G}/\mathcal{G}_{\text{CF}}$ needs an interval of $\mathcal{G}_{\text{CF}}$ excluding 0, and an MF personal-residual UAUC of at most 0.55 makes “no personal signal” a property of consumed-item panels.

6.2 What fine-tuning teaches

Table 4 separates what LoRA adds. Two item-only controls (Section 5) remove the user–item evidence: FT-C on ML-1M and FT-Q. By the wording rule of addendum 8, fine-tuning mostly teaches the item, and the adapter learns item quality from item text (FT-C on ML-1M ITEM_DRIVEN; FT-Q on ML-1M ITEM_DRIVEN, Toys ITEM_DRIVEN). Beside collaborative filtering, the fine-tuned

LLM’s prior and residual add to an MF stacker ($\mathcal{G}_{\text{LLM}|\text{CF}} > 0$, H-F) on 0 of 3 Qwen Amazon panels, and on sparse rows the item prior adds to the item mean ($\mathcal{G}_{\text{prior}} > 0$, H-S) in 0 of 2 zero-shot panel runs (Video Games below 150 users) and 0 of 2 fine-tuned panel runs (Video Games below 150 users); the other strata are descriptive. For store names, the Qwen3-8B knockout is labelled INDETERMINATE on Toys and Video Games in every model, so the confidence is called **[DATA NEEDED: branch: familiarity-linked / popularity-linked]**; the knockout replaces store names only, and the Llama Toys knockout is in Table 12. The popularity links (Table 3B) and the next-item popularity rows (Table 7B) are descriptive.

6.3 Which uses of the confidence survive a matched control

Sure when right, unsure when wrong (S1, S2). On **[DATA NEEDED: k of 4]** next-item domains acc_{high} is **[DATA NEEDED: above / indistinguishable from / below]** acc_{low} (difference **[DATA NEEDED: aud:C.acc_top_minus_bottom_tertile]**), so correct top-1 decisions are **[DATA NEEDED: concentrated in the high tertile / spread evenly / concentrated in the low tertile]**, and top-1 errors fall in the high tertile **[DATA NEEDED: more often than / as often as / less often than]** one third of the time (share **[DATA NEEDED: aud:C.share_errors_in_top_tertile]**; Table 7A). On rated panels Table 3C gives the margin’s AUROC and tertile shares for the oracle and the deployable top- k_u decision; only the deployable rows describe what a system could do.

Static exposure and selective serving (S3). For the next scores Δ_{head} is **[DATA NEEDED: above / indistinguishable from / below]** 0 on **[DATA NEEDED: k of 4]** domains (**[DATA NEEDED: aud:B.delta_head]**), with **[DATA NEEDED: the same / the opposite / no]** sign on the second backbone (**[DATA NEEDED: aud2l:B.delta_head]**), and **[DATA NEEDED: larger / similar / smaller]** than for the verbalised reranker and the eight published recommenders (Table 5). Serving the most confident half of the events against the random signal (Table 7C) is descriptive.

Corrections. Per-user maps change per-user AUC by at most 0.000, as Lemma 3.1 requires; the no-history and swap-prior corrections are descriptive (Table 10, rows 3–4), and the registered two-view contrast is labelled **[DATA NEEDED: mir:decision]** (rows 5–7).

Uncertainty-selected pruning (S6). The registered test asks whether pruning the TRAIN examples that the zero-shot model is most uncertain about (P2) helps more than random pruning matched per label class (P1) (Table 8). The contrast P2 – P1 is **[DATA NEEDED: prn:d_vs_P1]** (interval **[DATA NEEDED: prn:ci_P2_minus_P1]**), p **[DATA NEEDED: prn:p_P2_minus_P1]** with **[DATA NEEDED: prn:n_pos_of_5]** of 5 paired differences positive, so uncertainty-selected pruning is **[DATA NEEDED: better than / worse than / about equal to / inconclusive against]** class-matched random pruning for this signal, domain and backbone. As P1 matches label classes and not items, a win may reflect item composition; addendum 7 registers composition diagnostics (items left without examples, change of item label rates, TEST-item coverage per arm) and, only if P2 – P1 reads BEATS,

Table 3: What the confidence tracks: rated panels, Qwen3-8B (Llama: Table 12); ZS: zero-shot; LoRA: seed mean (s.d.); m : item mean, m_T : information-matched item mean; † : addendum 6 (Amazon: registered outcome-free; ML-1M, in italics: exploratory). P1: LoRA minus zero-shot $\mathcal{G}$ (seeds 0 / 1 / 2; mean, p). Estimators: Section 5.

	ML-1M		Toys		Video Games		Sports	
	ZS	LoRA	ZS	LoRA	ZS	LoRA	ZS	LoRA
<i>A. The LLM beside non-LLM scores on the same TEST rows</i>								
UAUC of ℓ	0.596±.022	0.742±.019 (.002)	0.527±.022	0.558±.020 (.002)	0.517±.022	0.580±.021 (.003)	0.493±.021	0.534±.020 (.005)
LoRA – ZS (E-B)		0.147±.022 (.002)		0.031±.025 (.002)		0.063±.026 (.003)		0.041±.027 (.005)
item mean m (ref.)		0.759±.021		0.684±.021		0.662±.022		0.624±.021
matched mean m_T (ref.) [†]		0.756		0.633		0.580		0.615
temporal MF (ref.)		0.791±.018		0.539±.021		0.532±.018		0.534±.022
$\Delta\text{UAUC}(\ell - m)$, E-J [†]	−0.163±.026	−0.016±.013 (.002)	−0.157±.030	−0.126±.028 (.002)	−0.144±.031	−0.082±.028 (.003)	−0.131±.031	−0.090±.029 (.005)
$\Delta\text{UAUC}(\ell - \text{MF})$, warm [†]	−0.195±.027	−0.048±.017 (.002)	−0.026±.039	−0.008±.038 (.001)	−0.073±.050	−0.007±.047 (.007)	−0.047±.037	−0.019±.034 (.006)
Users: TEST / S_d		366 / 366		943 / 943		820 / 820		997 / 997
<i>B. What the confidence is made of, and the information it adds (S_d)</i>								
Reliability r_8 of $\hat{\pi}$	0.773±.012	0.969±.002 (.002)	0.687±.020	0.800±.012 (.018)	0.624±.022	0.837±.011 (.031)	0.733±.016	0.762±.013 (.059)
Item-prior share ρ^2/r_8	0.393±.063	0.902±.012 (.001)	uninterpretable	0.662±.038 (.025)	uninterpretable	0.724±.027 (.033)	0.329±.047	0.741±.026 (.033)
e-share [†]	0.616±.069	0.098±.012 (.001)	0.736±.075	0.340±.037 (.027)	0.699±.047	0.283±.025 (.033)	0.702±.060	0.260±.025 (.033)
Item share of MF / label [†]		0.615±.033 / 0.213±.026		0.126±.030 / 0.085±.016		0.129±.036 / 0.075±.015		0.130±.028 / 0.059±.013
$\mathcal{G} = \Delta\text{UAUC}(M2 - M1)$, E-D	−0.002±.006	0.010±.007 (.001)	−0.002±.010	−0.005±.009 (.002)	−0.005±.007	−0.009±.008 (.001)	−0.009±.008	0.002±.009 (.002)
$\mathcal{G}_{\text{wu}}$ (E-W) [†]	0.000±.003	0.009±.007 (.001)	−0.003±.006	−0.003±.006 (.001)	−0.002±.002	−0.002±.002 (.000)	0.002±.003	0.001±.009 (.001)
$\mathcal{G}_{\text{CP}}$: E-D / E-W [†]		0.036±.013 / 0.037±.013		0.002±.005 / 0.001±.002		−0.002±.005 / −0.001±.002		0.002±.007 / 0.002±.004
MF personal residual (ref.)		0.653±.022		0.503±.028		0.519±.035		0.505±.025
Star permutation ΔUAUC	0.011±.008	0.009±.006 (.002)	0.008±.011	0.008±.008 (.004)	−0.002±.011	0.002±.009 (.002)	0.005±.010	−0.001±.007 (.003)
Popularity link of $\hat{\pi}$	0.368±.040	0.125±.048 (.032)	0.268±.023	0.254±.024 (.011)	0.098±.032	0.072±.030 (.025)	0.212±.025	0.027±.025 (.006)
Popularity link of ℓ	0.266±.040	0.148±.046 (.021)	0.061±.023	0.024±.023 (.008)	−0.014±.028	0.018±.028 (.022)	0.004±.023	−0.048±.023 (.009)
P1: $\mathcal{G}_{\text{TT}} - \mathcal{G}_{\text{ZS}}$	0.010 / 0.011 / 0.013; 0.011±.008, p 0.004			−		−		−
P1wu [†]	0.007 / 0.010 / 0.010; 0.009±.006, p 0.002			−		−		−
<i>C. Calibration and error anatomy of the per-user top-k_u decision (TEST rows)</i>								
ECE (after Platt)	0.020±.012	0.026±.009 (.002)	0.066±.015	0.055±.014 (.002)	0.058±.014	0.069±.015 (.003)	0.060±.014	0.056±.013 (.003)
Correct in bottom tertile, oracle	0.276±.009	0.264±.006 (.006)	0.290±.011	0.283±.008 (.008)	0.288±.011	0.282±.008 (.002)	0.289±.011	0.275±.007 (.003)
deployable [†]	0.294±.010	0.274±.006 (.002)	0.307±.011	0.295±.008 (.004)	0.304±.012	0.301±.008 (.008)	0.314±.011	0.292±.008 (.004)
Errors in top tertile, oracle	0.214±.018	0.133±.015 (.006)	0.259±.015	0.235±.014 (.005)	0.274±.015	0.237±.017 (.003)	0.256±.016	0.224±.014 (.008)
deployable [†]	0.267±.016	0.204±.015 (.008)	0.301±.014	0.274±.013 (.005)	0.299±.014	0.276±.014 (.001)	0.303±.014	0.266±.013 (.007)
Margin AUROC, oracle	0.633±.017	0.706±.014 (.008)	0.586±.016	0.609±.013 (.007)	0.579±.015	0.611±.015 (.004)	0.588±.016	0.627±.012 (.008)
deployable [†]	0.579±.016	0.643±.013 (.003)	0.543±.015	0.572±.013 (.003)	0.547±.015	0.566±.014 (.007)	0.535±.015	0.584±.013 (.008)

Table 4: What fine-tuning teaches: rated panels, Qwen3-8B; ZS: zero-shot; LoRA: seed mean (s.d.; knockout labels: seeds 0 / 1 / 2); † as in Table 3. C: LoRA gain retained under the item-only controls (registered, outcome-free). ML-1M has no store field.

	ML-1M		Toys		Video Games		Sports	
	ZS	LoRA	ZS	LoRA	ZS	LoRA	ZS	LoRA
<i>A. Complementarity in collaborative filtering (ℓ^2)</i>								
$\mathcal{G}_{\text{LUCY}} = \Delta\text{UAUC}(M1 - M2)$	0.004±.005 (.000)	−0.001±.006	0.000±.009 (.001)	−0.002±.004	0.003±.009 (.001)	0.000±.006	0.005±.012 (.002)	−0.001±.002 (.000)
$\mathcal{G}_{\text{CYLLM}} = \Delta\text{UAUC}(M1 - M2)$	0.017±.013	0.032±.012 (.001)	0.000±.002	−0.001±.001	0.000±.001 (.001)	−	−	−
<i>B. Where the item prior adds to the item mean ($\mathcal{G}_{\text{item}} = \Delta\text{UAUC}(M1 - M0)$ by stratum (E-F))</i>								
Square rows (in from < 5 values)	0.000±.000 (descriptive)	0.000±.000 (descriptive)	0.010±.015	−0.027±.033 (descriptive)	−0.011±.047 (.010) (descriptive)	0.013±.022	0.025±.032 (.009)	−0.001±.001 (.003)
Dense rows	−0.001±.001	0.000±.002 (.000)	0.000±.003	−0.002±.007 (.001)	0.000±.004	−0.001±.011 (.001)	0.001±.009 (.003)	0.001±.009 (.003)
Thinner items (no TRAIN example)	0.000±.000 (descriptive)	0.000±.000 (descriptive)	0.000±.000 (descriptive)	0.000±.000 (descriptive)	0.000±.000 (descriptive)	0.000±.000 (descriptive)	0.000±.000 (descriptive)	0.000±.000 (descriptive)
Seen items	0.000±.000	0.000±.002 (.000)	0.001±.004 (descriptive)	−0.025±.021 (.002) (descriptive)	0.000±.012	0.011±.025 (.002)	0.000±.016 (descriptive)	0.011±.037 (.005) (descriptive)
<i>C. Retention of the LoRA gain without user-item evidence</i>								
FTC (operational) R^2 reading	0.921±.050 / ITEM, DRIVEN	−	1.335±.136 / ITEM, DRIVEN	−	[DATA NEEDED: exFT_Q_reading_R] / [DATA NEEDED: exFT_Q_reading_label]	−	[DATA NEEDED: exFT_Q_reading_R] / [DATA NEEDED: exFT_Q_reading_label]	−
FTQ (insider) R^2 reading	0.851±.087 / ITEM, DRIVEN	−	0.561 / 0.371 / 0.484	−	[DATA NEEDED: exFT_Q_reading_LIAC]	−	[DATA NEEDED: exFT_Q_reading_LIAC]	−
UAUC of q_1 , 1m	0.720 / 0.720 / 0.720	−	−	−	−	−	−	−
<i>D. Store name knockout (pseudonym arm placebo real store swap)</i>								
Head – tail drop of ℓ	−	0.400±.129	0.007 (.011)	−0.542±.308	0.107 (.010)	0.161±.121	0.011 (.006)	0.011 (.006)
Florida drop	−	0.180±.141	0.040 (.010)	−0.309±.130	0.101 (.023)	−0.010±.131	0.006 (.003)	0.006 (.003)
Tail ALIAC [†]	−	0.010±.015	0.013 (.001)	0.004±.010	0.001 (.000)	0.009±.015	0.020 (.003)	0.020 (.003)
Registered label	−	−	INDETERMINATE / INDETERMINATE / INDETERMINATE	INDETERMINATE	INDETERMINATE / INDETERMINATE / INDETERMINATE	INDETERMINATE	INDETERMINATE / INDETERMINATE / INDETERMINATE	INDETERMINATE

Table 5: Static exposure of the top-10 lists (S3); LLM: Qwen3-8B, V0; the verbalised reranker and the eight published recommenders are context (min, median and max over the eight; Gini, APLT and min/median/max: estimates).

Method	Aloud				Gini of exposure				APLT			
	S3k	S3k	Toys	Home	S3k	S3k	Toys	Home	S3k	S3k	Toys	Home
Real (dev)	0.084±.001	0.089±.001	0.082±.001	[DATA NEEDED: exR_posl, head, share]	[DATA NEEDED: exR_posl, head, share]	0.099±.007	0.090±.007	[DATA NEEDED: exR_posl, tail, share]	[DATA NEEDED: exR_posl, tail, share]	[DATA NEEDED: exR_posl, tail, share]	[DATA NEEDED: exR_posl, tail, share]	[DATA NEEDED: exR_posl, tail, share]
LLM (dev)	0.057±.000	0.059±.001	0.059±.001	[DATA NEEDED: exR_delta, head]	[DATA NEEDED: exR_delta, head]	0.074±.004	0.065±.005	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]
LLM (dev)	0.055±.000	0.057±.001	0.052±.001	[DATA NEEDED: exR_delta, head]	[DATA NEEDED: exR_delta, head]	0.075±.004	0.060±.005	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]
Verbalised reranker	0.089±.000	0.085±.001	0.075±.001	[DATA NEEDED: exR_delta, head]	[DATA NEEDED: exR_delta, head]	0.072±.000	0.065±.005	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]	[DATA NEEDED: exR_delta, tail, share]
Published, min	0.100	0.100	0.095	[DATA NEEDED: exR_delta, head, min of 8]	[DATA NEEDED: exR_delta, head, min of 8]	0.074±.000	0.060±.005	[DATA NEEDED: exR_delta, tail, share, min of 8]	[DATA NEEDED: exR_delta, tail, share, min of 8]	[DATA NEEDED: exR_delta, tail, share, min of 8]	[DATA NEEDED: exR_delta, tail, share, min of 8]	[DATA NEEDED: exR_delta, tail, share, min of 8]
Published, median	0.264	0.263	0.267	[DATA NEEDED: exR_delta, head, median of 8]	[DATA NEEDED: exR_delta, head, median of 8]	0.085±.000	0.060±.005	[DATA NEEDED: exR_delta, tail, share, median of 8]	[DATA NEEDED: exR_delta, tail, share, median of 8]	[DATA NEEDED: exR_delta, tail, share, median of 8]	[DATA NEEDED: exR_delta, tail, share, median of 8]	[DATA NEEDED: exR_delta, tail, share, median of 8]
Published, max	0.266	0.263	0.266	[DATA NEEDED: exR_delta, head, max of 8]	[DATA NEEDED: exR_delta, head, max of 8]	0.095±.000	0.060±.005	[DATA NEEDED: exR_delta, tail, share, max of 8]	[DATA NEEDED: exR_delta, tail, share, max of 8]	[DATA NEEDED: exR_delta, tail, share, max of 8]	[DATA NEEDED: exR_delta, tail, share, max of 8]	[DATA NEEDED: exR_delta, tail, share, max of 8]

an item-stratified random arm: **[DATA NEEDED: branch: their outcome]**.

Optional method (Section 5; Table 11). Prior-offset LoRA is **[DATA NEEDED: above / indistinguishable from / below]** post-hoc stacking on **[DATA NEEDED: k of 4]** datasets (gain **[DATA NEEDED: slot:d_offset_minus_stack]**), so the method is **[DATA NEEDED: branch: retained / retired and reported in the appendix]**.

7 Conclusion

We audited what the token confidence of yes/no LLM recommenders encodes, with matched labels, non-LLM references on the same pairs and a protocol registered before any confirmatory result was analysed. Splitting the confidence into a user offset, an item prior and a pair-specific residual is an accounting identity, and the invariance lemma is a property of within-user metrics; what they leave open was measured. What the confidence knows is mostly the item. The fine-tuned LLM ranks within users below an item mean computed on the same rows on all four rated panels, the item prior

Table 6: One-table guide to S1–S6, from the tables above only (registered endpoints: Table 2; same-candidate scope). “Admitted” applies the claim-admission rule of Section 4; a row not admitted stays direction-free.

Seed question (tables)	Answer	Actionable via (control)	Admitted
S1 Is the model sure when it is right? (Tables 7, 3C)	[DATA NEEDED: one clause from Tables 7, 3C]	[DATA NEEDED: cross-user serving vs. random signal (Table 7C)]	[DATA NEEDED: admitted: yes / no]
S2 When it is wrong, was it unsure? (Tables 7, 3C)	[DATA NEEDED: one clause from Tables 7, 3C]	[DATA NEEDED: selective serving (Table 7C); not per-user maps (Lemma 3.1)]	[DATA NEEDED: admitted: yes / no]
S3 Does confident serving over-expose head items? (Tables 5, 7C, 14)	[DATA NEEDED: one clause from Tables 5, 7C]	[DATA NEEDED: selective serving vs. random signal: niche served share (Table 7C)]	[DATA NEEDED: admitted: yes / no]
S4 What does confidence track, and what does fine-tuning teach? (Tables 3, 4)	[DATA NEEDED: one clause from Tables 3, 4]	[DATA NEEDED: item-dependent corrections vs. controls (Table 10)]	[DATA NEEDED: admitted: yes / no]
S5 Popular items more confident, and warranted? (Tables 7B, 3B, 4D)	[DATA NEEDED: one clause from Tables 7B, 3B, 4D]	[DATA NEEDED: store-name knockout vs. real-store placebo (Table 4D)]	[DATA NEEDED: admitted: yes / no]
S6 Does uncertainty-selected pruning help more than class-matched random pruning? (Table 8)	[DATA NEEDED: one clause from Table 8]	[DATA NEEDED: training-time only; control: random pruning P1, class-matched (Table 8)]	[DATA NEEDED: admitted: yes / no]

Table 7: Next-item audit, Qwen3-8B zero-shot, V0, next; S1k, S10k: Sports events 1–1000, 1001–10000. C: serving at 50% coverage (the comparison with the baselines’ own confidence is not run: no score files).

	S1k	S10k	Toys	Home	Toads
A. Error anatomy: calibration, quality	11.543	11.003	9.766	[DATA NEEDED: andC.temperature]	[DATA NEEDED: andC.temperature]
Temperature T	0.021 ± 0.014	0.019 ± 0.006	0.040 ± 0.006	[DATA NEEDED: andC.avg]	[DATA NEEDED: andC.avg]
ECE (top label)	0.072 ± 0.004	0.072 ± 0.001	0.050 ± 0.002	[DATA NEEDED: andC.head]	[DATA NEEDED: andC.head]
Brier	0.048 ± 0.023	0.039 ± 0.007	0.045 ± 0.007	[DATA NEEDED: andC.misrate, correct, rate]	[DATA NEEDED: andC.misrate, correct, rate]
ooblogp	0.180 ± 0.042	0.157 ± 0.113	0.294 ± 0.115	[DATA NEEDED: andC.tier0, accuracy]	[DATA NEEDED: andC.tier0, accuracy]
Errors on high tier0	0.304 ± 0.029	0.308 ± 0.010	0.274 ± 0.009	[DATA NEEDED: andC.share, errors, in, top, tier0]	[DATA NEEDED: andC.share, errors, in, top, tier0]
ADBCQ (Open)	0.770 ± 0.061	0.752 ± 0.021	0.766 ± 0.014	[DATA NEEDED: andC.prior, p, mean, top1]	[DATA NEEDED: andC.prior, p, mean, top1]
ADBCQ (P) pooled	0.780 ± 0.016	0.712 ± 0.005	0.719 ± 0.005	[DATA NEEDED: andC.prior, p, mean, top1]	[DATA NEEDED: andC.prior, p, mean, top1]
UAUC (per event)	0.790 ± 0.017	0.751 ± 0.006	0.724 ± 0.005	[DATA NEEDED: andC.prior, p, mean, top1]	[DATA NEEDED: andC.prior, p, mean, top1]
NDCG@10	0.299 ± 0.019	0.297 ± 0.007	0.245 ± 0.007	[DATA NEEDED: andC.random, andC]	[DATA NEEDED: andC.random, andC]
P. Popularity	0.001 ± 0.001	0.001 ± 0.001	−0.002 ± 0.001	[DATA NEEDED: andC.head, min, tail, mean, p]	[DATA NEEDED: andC.head, min, tail, mean, p]
Mean p, head + tail	−0.012 ± 0.014	−0.001 ± 0.005	−0.004 ± 0.006	[DATA NEEDED: andC.head, min, tail, mean, p]	[DATA NEEDED: andC.head, min, tail, mean, p]
Star index, head + tail	0.753 ± 0.026	0.738 ± 0.009	0.727 ± 0.009	[DATA NEEDED: andC.head, min, tail, mean, p]	[DATA NEEDED: andC.head, min, tail, mean, p]
Top-1 in head (share)				[DATA NEEDED: andC.head, min, tail, mean, p]	[DATA NEEDED: andC.head, min, tail, mean, p]
C. Selective serving	0.034	0.062	0.103	[DATA NEEDED: andC.gain, at, 30, vs, full]	[DATA NEEDED: andC.gain, at, 30, vs, full]
NDCG@10 gain, random	−0.001	−0.006	−0.007	[DATA NEEDED: andC.gain, at, 30, vs, full]	[DATA NEEDED: andC.gain, at, 30, vs, full]
NDCG@10 gain, random	−0.001	−0.002	0.144	[DATA NEEDED: andC.gain, at, 30, vs, full]	[DATA NEEDED: andC.gain, at, 30, vs, full]
Niche = mainstream, Pavg	0.004	−0.002	0.017	[DATA NEEDED: andC.niche, min, mainstream, served, share]	[DATA NEEDED: andC.niche, min, mainstream, served, share]
Niche = mainstream, random				[DATA NEEDED: andC.niche, min, mainstream, served, share]	[DATA NEEDED: andC.niche, min, mainstream, served, share]

Table 8: Pruning (S6), ML-1M, Qwen3-8B LoRA, TEST UAUC: mean (seed s.d.); each pruned arm removes 25% of each label class; Δ: seed-averaged contrast; > 0: positive paired differences of 5.

Arm	UAUC	Δ vs. P1	> 0	Δ vs. P0
P0 full data	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]
P1 random (class-matched)	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]
P2 uncertainty-selected	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]
P3 prior-congruent	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]	[DATA NEEDED: pruned UAUC, mean, sd]
Item mean, popularity, MF (ref.)	0.759 ± 0.021 / 0.616 ± 0.020 / 0.791 ± 0.018			

explains 0.66 to 0.90 of the within-user variance of its confidence, and the pair-specific residual adds at most 0.010 UAUC to the prior and the item mean (ML-1M with Qwen3-8B only; not replicated with Llama-3.1-8B, and confirmed negative on one zero-shot panel), nothing detectable to matrix factorisation (confirmed on none of the three Amazon panels). Fine-tuning mostly teaches the item: two item-only controls keep 0.85 to 0.92 of its gain on ML-1M, and the item-only teacher reaches the real-label adapters’ UAUC on Toys. Each route that the lemma leaves open was tested against its own control. [DATA NEEDED: 1–2 sentences from the tables of Section 6.3 only: which of item-dependent corrections, selective serving, static exposure and uncertainty-selected pruning beat their placebo, ensemble null, random or class-matched random control, with effect size and interval, or that none

did] [DATA NEEDED: branch: only if the nested method slot (prior-offset LoRA) survives its kill rule – one sentence on its gain over post-hoc stacking; else omit] Both registered gates were met on ML-1M: the remedied zero-shot prompt reached UAUC 0.603 on untouched users (bar 0.60) and LoRA tuning reached 0.742 (bar 0.65), so the fine-tuned regime carries the main results and the zero-shot regime is the contrast. Table 6 condenses the answers to the six questions. [DATA NEEDED: one sentence of practitioner guidance drawn only from the admitted rows of that table]

Limitations. One LoRA recipe (rank 16, one epoch) on two 8B backbones is audited, and no sampling-based or verbalised uncertainty estimator. Scores are pointwise, the next-item panels hold sampled candidates under same-candidate scope [35], and exposure is static. The knockout replaces store names only, and its registered thresholds are on the logit scale, hence unit-dependent. The item-only controls are partial: permutation informs on ML-1M only (97.7% of its TRAIN examples sit in repeated items), and the FT-Q teacher, set by the item mean, departs from the real labels, so a high R_Q shows what an item-level target can teach, not what the real labels taught. The ML-1M values of the addendum-6 endpoints are exploratory, and item text, ML-1M included, may have been seen in pre-training [11]. The registration is a set of dated, hash-chained project documents rather than a public-registry entry, and the reviews behind addenda 6 and 7 were internal, not independent.

A Prompts, Registration and Reproducibility

All prompts are single-turn chat messages (thinking disabled). V0, the registered prompt of the rated panels (user message, no system message):

```
You are an expert recommendation system.
User history (oldest to newest):
- {title} (rated {r}/5) [last 10 events, one line each]
Candidate item:
Title: {title}
Description: {first 200 characters, if present}
```

Would this user like the candidate item? Answer with only Yes or No.

Table 9: Deviations, post-hoc choices and disclosures, each checkable in the dated pilot log (log), the amendments (Ak; A3-k: addendum k of A3) or the commit history.

	What happened (evidence)	Effect on claims
1	A method paper became an audit when the Pilot-1 gate failed ($0.587 < 0.60$); A2 then closed a gap in its decision table (§3.1).	The audit design postdates the first data.
2	P1 and E-D were written after exploratory forensics on the burned pilot users (log 10-04).	Only P1's fine-tuned side is confirmatory.
3	The stage-3 ensemble null is the code's binormal prediction, not A2 G7's literal z-sum (logged before stage 3).	MIRKOR's null follows the code.
4	The wording rules were written after the ML-1M results, hashed before any later file was pulled (A3-6 §9.3).	The ML-1M wording was set with its numbers known.
5	A comment rule allowed direction words on unadjusted intervals; never implemented (A3-6 §1.2).	None.
6	The Gate-FT bar is anchored to published ML-1M UAU/Cs with 3-star negatives (A2 G9).	It licenses the programme, not competence.
7	V1 was selected on burned DEV users (also TRAIN users): DEV 0.612, CONFIRM 0.603, zero-shot on TEST rows 0.596, against the 0.60 bar applied to CONFIRM users.	Gate met on the point estimate; zero-shot rated results are scored to V1.
8	Which arm a GPU-hour cut drops is chosen with ML-1M results in view (A3 §10).	Each cut is logged before the affected result is read.
9	An INCOMPLETE method-slot dataset needs an addendum (A3-5).	Method slot only.
10	The niche and mainstream bins were redefined after the Sports analysis, before any other niche value (spec G).	Not quintiles; group sizes vary by domain.
11	The next-item specification was written while scoring ran and frozen before any analysis (log 10-03/04).	None known.
12	Amendment 1's hash was logged retroactively, from commit and pull times (log 10-04).	A1's timing rests on those times.
13	The non-prior share was called an upper bound on the personal share (A3); withdrawn (A3-6 §1.1).	$\mathcal{G}$ measures personal information.
14	E-D's stackers are pooled and uncentred; E-W was added after a review (A3-6 §2).	Disagreement: estimator-dependent.
15	Addendum 6 was written after the ML-1M report and two internal reviews (log 10-04).	Its ML-1M values ($\dagger$) are exploratory.
16	m may use ratings in $\{T_d, t\}$, which the LLMs never see (A3-1 item 2; A3-6 §6).	m_T is printed beside it.
17	E-C's top- k_u takes k_u from the TEST likes, an oracle (A3-6 §7).	Only deployable rows describe a system.
18	Gate-FT's file holds a second scoring of the ML-1M zero-shot prompts (vLLM batching).	None: the grid scoring is printed.
19	A3-6 added Toys to FT-C (5.1% of TRAIN labels can move); A3-8 withdrew it before any control adapter and added FT-Q.	FT-C: ML-1M only; the wording needs FT-Q too.
20	Code readings: R needs a complete E-B and 150 users; a missing member keeps $p = 1$; k_u rounds half up; E-W needs AUCs in all 20 splits.	No threshold or family changes.

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Correlation (control)	ML-IM	Inf-Inf	Spots
8 Raw TAUIC (item mean, 10P test)	0.9964/02/17961032/0.97641035	[DATA NEEDED rawTAUIC_item_mean_test_TAUIC_inf_inf_temporal]	-
1 Per user TAUIC rating (sum, 10P test)	0.000	[DATA NEEDED rawTAUIC_sum_rdc_AUC_inf_inf]	-
10 User rate (last training, 3-point AUC)	-0.010	[DATA NEEDED user_rate_inf_inf_posited_inf_inf_temporal]	-
10 No history per rating (sum, 10P test)	-0.9364/04	[DATA NEEDED rawTAUIC_no_history_inf_inf]	-
10 User rate (inf-Inf, 10P test)	0.0094/043	[DATA NEEDED user_rate_inf_inf]	-
10 Mean (pageview per phase)	[DATA NEEDED rawTAUIC_mean_inf_inf_posited]	[DATA NEEDED rawTAUIC_mean_inf_inf]	-
10 Mean (clickable rate)	[DATA NEEDED rawTAUIC_mean_inf_inf_clickable]	[DATA NEEDED rawTAUIC_mean_inf_inf_clickable]	-
10 Mean (conversion rate)	[DATA NEEDED rawTAUIC_mean_inf_inf_conversion]	[DATA NEEDED rawTAUIC_mean_inf_inf_conversion]	-
10 Mean (conversion rate, sum, 10P test)	0.8095	[DATA NEEDED rawTAUIC_mean_inf_inf_conversion_sum]	-

	IL-IM	Topo	Video Games	Sports
SPT, $k = 0$ (LaRA of the mean graph [%])	0.7424-0.019 (002)	0.7504-0.009 (002)	0.7564-0.021 (003)	0.7544-0.030 (005)
Post-hoc stacking	[DATA NEEDED: slot U(AUC_mean_sd)]	[DATA NEEDED: slot U(AUC_mean_sd)]	[DATA NEEDED: slot U(AUC_mean_sd)]	[DATA NEEDED: slot U(AUC_mean_sd)]
Prior-offer LaRA	[DATA NEEDED: slot off(AUC_mean_sd)]	[DATA NEEDED: slot off(AUC_mean_sd)]	[DATA NEEDED: slot off(AUC_mean_sd)]	[DATA NEEDED: slot off(AUC_mean_sd)]
Offset - stacking	[DATA NEEDED: slot off(AUC_mean_sd)]	[DATA NEEDED: slot off(AUC_mean_sd)]	[DATA NEEDED: slot off(AUC_mean_sd)]	[DATA NEEDED: slot off(AUC_mean_sd)]
Item mean / MF (out)	0.7794-0.021 / 0.7914-0.01	0.6840-0.021 / 0.5794-0.021	0.6424-0.021 / 0.5324-0.03	0.6244-0.021 / 0.5344-0.02

	ML-1M		Toys	
	ZS	LoRA	ZS	LoRA
UAUC of $\mathcal{D}$	0.605±0.022	0.760±0.018 (0.001)	0.512±0.022	0.560±0.020 (0.004)
UAUC of $\mathcal{D}$ (E-R)	-	0.154±0.025 (0.001)	-	0.048±0.023 (0.004)
Item mean μ (ref.)	-	0.759±0.021	-	0.684±0.021
temporal μ (ref.)	-	0.791±0.018	-	0.539±0.021
UAUC of $\mathcal{D}$ - μ , $E^{\dagger}$	-0.152±0.029	0.600±0.015 (0.001)	-0.171±0.030	-0.123±0.026 (0.004)
ECE (after Platt)	0.034±0.015	0.020±0.010 (0.000)	0.060±0.014	0.052±0.013 (0.001)
Cross in bottom tercile, orcale	0.272±0.009	0.263±0.006 (0.003)	0.289±0.011	0.280±0.007 (0.002)
Errors in top tercile, orcale	0.229±0.018	0.127±0.015 (0.003)	0.254±0.015	0.277±0.015 (0.003)
Item-prior share ρ^2/r_0	uninterpretable	0.860±0.020 (0.014)	uninterpretable	0.810±0.021 (0.012)
e-share [†]	0.748±0.065	0.40±0.020 (0.014)	0.733±0.116	0.190±0.021 (0.012)
$\mathcal{G}_W$ = MAUC (M2-M1), E-D	0.003±0.006	0.070±0.008 (0.001)	0.003±0.009	-0.002±0.010 (0.001)
$\mathcal{G}_W$ = EAUUC	0.003±0.004	0.002±0.007 (0.001)	0.000±0.005	-0.002±0.006 (0.001)
$\mathcal{G}_{CT}$ = $\mathcal{G}_W$, $E^{\dagger}$, $E^{\dagger}\mu^{\dagger}$	0.036±0.013	0.010±0.013 (0.001)	-	0.001±0.002
Popularity limit of $\hat{\pi}$	0.329±0.040	0.017±0.048 (0.036)	0.264±0.024	0.224±0.024 (0.014)
Π : $\hat{\mathcal{G}}_{\Pi}$ - $\hat{\mathcal{G}}_Z$	0.005 / 0.003 / 0.004	0.04±0.008, ρ 0.382	-	-
Π_{new}	0.006 / 0.008 / 0.006	0.007±0.007, ρ 0.054	-	-
Knock-out: head - tail drop of $\hat{\pi}$	-	-	0.054±0.026	0.079±0.027
Knock-out: registered label	-	-	INDETERMINATE	INDETERMINATE / INDETERMINATE

	UAUC (seeds 0 / 1 / 2)	All rows (seeds 0 / 1 / 2)	Item mean (ref.)
ML-1M	0.740 / 0.745 / 0.743	0.733 / 0.734 / 0.734	0.759±.021
Toys	0.559 / 0.556 / 0.559	0.579 / 0.578 / 0.581	0.684±.021
Video Games	0.584 / 0.578 / 0.579	0.607 / 0.606 / 0.611	0.662±.021
Sports	0.537 / 0.537 / 0.529	0.558 / 0.549 / 0.554	0.624±.021
ML-1M, permuted labels (seeds 0 / 1)	0.731 / 0.730	0.721 / 0.722	–
ML-1M, real – permuted (seeds 0 / 1)	0.009±.008 / 0.014±.008	0.012±.003 / 0.012±.003	–

[illegible]

Reproducibility checklist. Scoring: vLLM prefill, at most 4,096 tokens (3,072 next-item), 100-user chunks. *Fine-tuning:* LoRA [28] on the query, key, value and output projections, $r=16$, $\alpha=32$, dropout

0.05, learning rate 10^{-4} , cosine decay, 3% warm-up, one epoch, bf16, effective batch 32; at most 24,000 training examples per domain; prompts above 1,024 tokens skipped and counted; FT-C and FT-Q controls: seeds 0 and 1. **Data and hashes:** pilot ML-1M panel 985494c7, pilot Toys panel 69252b48, registered V0 prompts on the pilot ML-1M panel 12e83c4f (first eight hexadecimal digits of the sha1); **[DATA NEEDED: amendment sha1s, rendered prompt-bank sha1, user-id list sha1s and the time split T_d of every domain from the freeze record and the split files].** **Software, compute, artefact:** **[DATA NEEDED: package versions and GPU model from each report; total GPU hours].** Prompts, panel builders, scorers, analysis scripts and every result file are released with the anonymous artefact.

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